

Exp no :09 Using Process Explorer to Identify Suspicious Processes

Date :

Description

- Process Explorer is a part of the Microsoft Sysinternals Suite. It is a powerful tool used to view detailed information about system processes.
- It helps investigators and administrators analyze active processes, detect suspicious behavior, monitor CPU and memory usage, and verify process authenticity using digital signatures.

Step-by-Step Process

Step 1: Download and Setup Process Explorer

1. Go to the official Microsoft Sysinternals website: <https://learn.microsoft.com/en-us/sysinternals/downloads/process-explorer>
2. Click Download Process Explorer.
3. Extract the downloaded ZIP file to a folder.
4. Right-click procexp64.exe (for 64-bit) or procexp.exe (for 32-bit) → select Run as Administrator.

Step 2: Understand the Interface

1. The main window displays all running processes in a hierarchical tree view.
2. Each process shows details such as PID, CPU usage, memory usage, and company name.
3. Color codes represent process states:
 - Green — Newly started processes
 - Red — Terminated processes
 - Light Blue — Processes running under the current user
 - Pink — Suspended processes

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Secure System		172 K	64,896 K	188		
Registry	0.12	8,184 K	54,140 K	228		
System Idle Process	74.01	60 K	8 K	0		
System	1.67	68 K	11,360 K	4		
Interrupts	1.31	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,160 K	1,420 K	888		
Memory Compression	< 0.01	2,404 K	10,02,652 K	3956		
csrss.exe		2,572 K	6,728 K	1276		
wininit.exe		1,732 K	8,872 K	1364		
services.exe	0.83	7,808 K	20,904 K	1512		
svchost.exe	< 0.01	15,000 K	44,172 K	1672	Host Process for Windows S...	Microsoft Corporation
WmiPrvSE.exe		4,656 K	14,888 K	7204	WMI Provider Host	Microsoft Corporation
WmiPrvSE.exe	7.73	23,556 K	31,628 K	7212	WMI Provider Host	Microsoft Corporation
unsecapp.exe		2,328 K	11,108 K	7752	Sink to receive asynchronous...	Microsoft Corporation
StartMenuExperienceHos...	< 0.01	1,44,188 K	1,66,744 K	11404	Windows Start Experience H...	Microsoft Corporation
SearchHost.exe		86,132 K	1,36,876 K	11412		Microsoft Corporation
msedgewebview2.exe	< 0.01	43,944 K	1,16,864 K	13064	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		3,000 K	12,640 K	13168	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		15,556 K	41,260 K	13456	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		1,21,776 K	81,208 K	13472	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		9,804 K	18,396 K	13660	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		1,37,472 K	1,42,564 K	13836	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		8,868 K	20,556 K	17636	Microsoft Edge WebView2	Microsoft Corporation
WidgetBoard.exe		1,15,112 K	1,14,396 K	11668		Microsoft Corporation
RuntimeBroker.exe		19,436 K	77,324 K	11824	Runtime Broker	Microsoft Corporation
WidgetService.exe		5,924 K	29,100 K	6976	WidgetService.exe	Microsoft Corporation
unsecapp.exe		1,668 K	9,980 K	11864	Sink to receive asynchronous...	Microsoft Corporation
MicrosoftStartFeedProvid...		10,500 K	42,812 K	15304		Microsoft Corporation
RuntimeBroker.exe		2,436 K	14,168 K	15792	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		3,248 K	22,392 K	15784	Runtime Broker	Microsoft Corporation
TextInputHost.exe	0.36	2,32,896 K	2,72,072 K	15700		Microsoft Corporation
RuntimeBroker.exe		2,356 K	13,284 K	12512	Runtime Broker	Microsoft Corporation
IGCC.exe		44,144 K	59,684 K	8396	IGCC	Intel Corporation
secd.exe		4,560 K	19,504 K	18924	Apple Security Manager	Apple, Inc.
APSDaemon.exe		6,116 K	25,056 K	17304	Apple Push	Apple Inc.
ApplePhotoStreams.exe		8,932 K	41,740 K	21376	iCloud Photo Stream	Apple Inc.
ShellExperienceHost.exe	Susp...	62,008 K	87,020 K	22396	Windows Shell Experience H...	Microsoft Corporation
RuntimeBroker.exe		8,116 K	42,460 K	20896	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		2,456 K	14,232 K	2196	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		6,532 K	30,472 K	21816	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		8,036 K	26,006 K	21668	Runtime Broker	Microsoft Corporation

CPU Usage: 26.41% Commit Charge: 58.66% Processes: 319 Physical Usage: 65.24%

Step 3: Identify Suspicious Processes

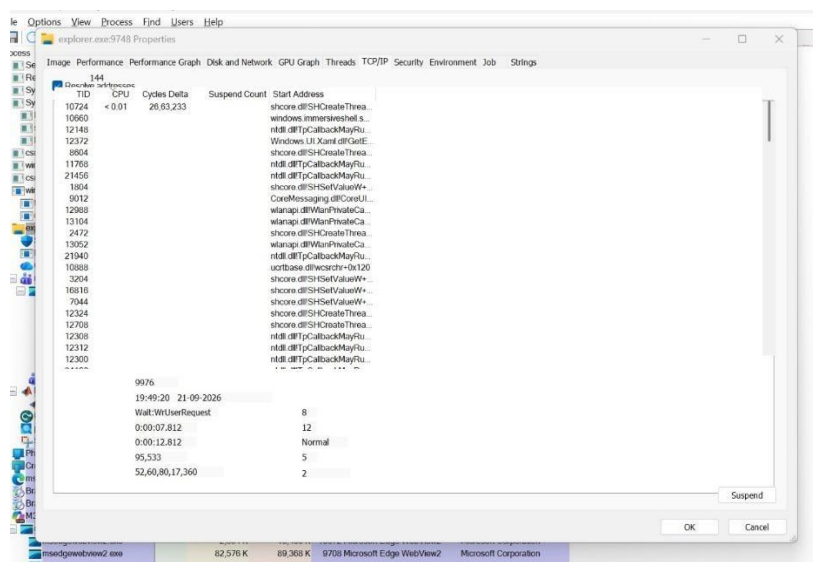
1. Look for unfamiliar or oddly named processes (e.g., xkdjeo.exe, randomname123.exe).
2. Check the Company Name and Description:
 - o Legitimate software usually shows known publishers like *Microsoft, Intel, or Adobe*.
3. Right-click the process → Properties → go to the Image tab.
4. Verify the Path of the executable file:
 - o Safe: C:\Windows\System32\ o Suspicious: C:\Users\<User>\AppData\Temp\ or Downloads\
5. Check for Digital Signature:
 - o Valid signature = trusted developer o No signature or invalid = possibly malicious

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Step 4: Analyze Process Behavior

- Observe CPU, Memory, and I/O usage columns.
- If a small or unknown process consumes excessive CPU or memory, it may be malicious.
- Right-click the process → Properties → go to the TCP/IP tab.
- Check if it communicates with unknown external IP addresses.
- Examine Handles and DLLs tabs for suspicious loaded files or libraries.

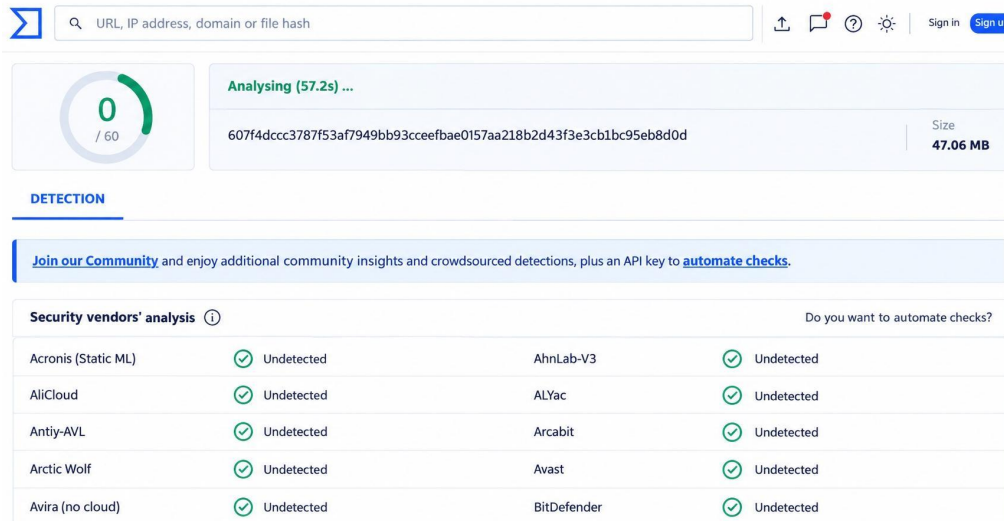


Step 5: Verify Process Legitimacy

1. Search the process name on Google.

Example: svchost.exe vs svhst.exe (one letter missing — suspicious).

2. Visit <https://www.virustotal.com> o Upload the process file or search its name to verify if it's reported as malware.
3. Cross-check with ProcessLibrary.com or official vendor websites for authenticity.



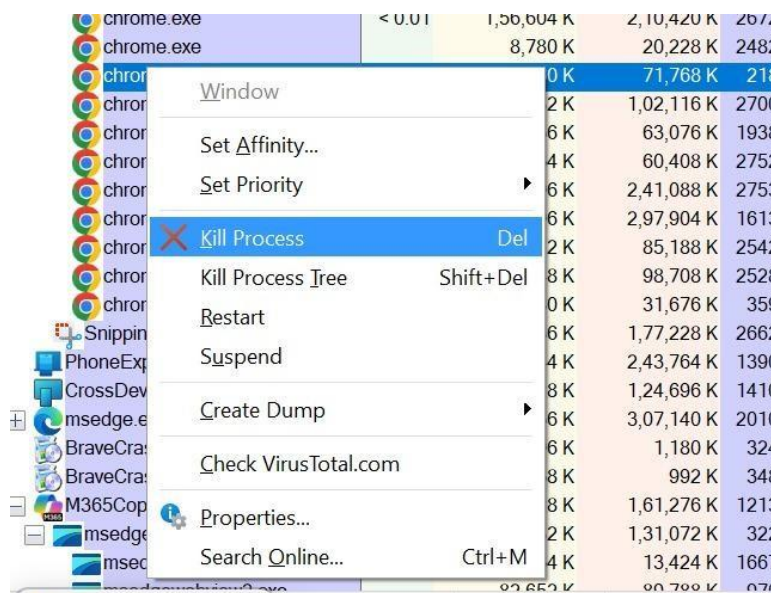
The screenshot shows the VirusTotal web interface. At the top, there's a search bar with the text "URL, IP address, domain or file hash". Below it, a progress indicator shows "Analysing (57.2s) ...". A file hash is displayed: "607f4dccc3787f53af7949bb93ccefbae0157aa218b2d43f3e3cb1bc95eb8d0d". The file size is "47.06 MB".

Under the "DETECTION" tab, there's a section for "Security vendors' analysis". It includes a table of results from various vendors, all showing "Undetected".

Security vendors' analysis	Do you want to automate checks?
Acronis (Static ML)	Undetected
AliCloud	Undetected
Antiy-AVL	Undetected
Arctic Wolf	Undetected
Avira (no cloud)	Undetected
AhnLab-V3	Undetected
ALYac	Undetected
Arcabit	Undetected
Avast	Undetected
BitDefender	Undetected

Step 6: Take Appropriate Action

1. **If the process is confirmed malicious:**
 - o Right-click the process → Kill Process to stop it.
 - o Delete the corresponding executable file from its path.
2. **If unsure:** o Right-click → Suspend Process to stop it temporarily for investigation.
3. **After removal:**
 - o Run a Full System **Scan** using Windows Defender or Malwarebytes to ensure no remnants remain.



Step 7: Example Observation

You find faangpath_simple_template.pdf consuming 70% CPU.

- **Path:** C:\Users\Admin\AppData\Temp\faangpath_simple_template.pdf
- **Digital Signature:** None
- **Company Name:** Unknown
- **Network Activity:** Shows connections to unknown IPs in the TCP/IP tab
- **Online Check:** VirusTotal confirms it as a known trojan
- **Action Taken:** Suspended → Killed → Deleted file → Performed full antivirus scan



Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result

Using Process Explorer, suspicious processes were successfully identified by examining their CPU usage, path, digital signature, and network activity.

Confirmed malicious processes were terminated and removed to maintain system integrity.

